Weekly Report: Ultra-High-Pressure ODMR — How much of the predicted multiplicity ladder can we actually measure?

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Question

> Above the splitting power I_c the optimum is a degenerate level set, so what an experiment actually sees is the η bump *between* its members rather than the members themselves — how many of the seven predicted rungs survive that, and what power grid do the survivors demand?

Hypothesis

If members closer than the addressable resolution (0.5 nm) merge and a rung counts as read only when every bump separating its members clears $3\sigma_\eta$, then what limits the experiment is bump *depth* rather than rung width, and the power grid is fixed by the narrowest *readable* rung instead of the narrowest one.

Evidence / Interpretation

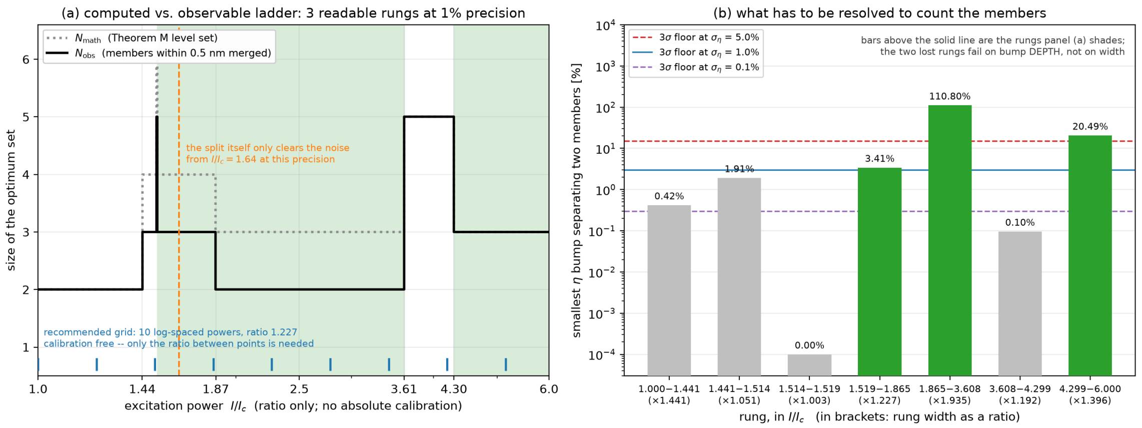

図 1: (a) The ladder as Theorem M computes it (N_{math}) and as an experiment sees it (N_{obs}, members within 0.5 nm merged); shading marks the rungs readable at $\sigma_\eta = 1\%$, ticks the recommended ten-point grid. (b) The smallest η bump on each rung against the 3σ detection floor; bar labels give the bump, brackets the rung width.

- **Depth, not width, decides what is measurable**: the smallest bump on a rung ranges over four orders of magnitude (0.00%–110.80%), so at $\sigma_\eta = 1\%$ only three of the seven rungs are readable — the $\times 1.192$-wide five-member rung is lost to depth (0.10%), and the split itself, opening from zero at I_c, does not clear 3σ until $I/I_c \approx 1.64$ — which fixes the experiment as **ten log-spaced powers of ratio** 1.227 **from** $I/I_c = 1$ **to** 6, calibration free, aimed at the deep transitions at 1.865 and 4.299 rather than at the split.

Gap / Failure

- **Gap** — the bump depths that decide every verdict above are computed from the Ho Fig. 1(e) kernel reconstruction (L2, kernel conditional), unlike the level-set structure itself, and I_c has never been located experimentally, so the required span $I_{\max}/I_c \geq 4.3$ cannot yet be checked against our source.
- **Failure** — the headline claim, "the optimum divides," is retired as a target: it needs sub-0.1% photometry, and this week's calculation replaces it with the multiplicity transitions.

Next Step

Anchor I_c from a coarse scan on the robust $I/I_c = 1.865$ transition (not on the split onset, which is below the 1% floor), then run the ten-point ratio-1.227 sweep at 1% photometry while recording the NV^0/NV^- ratio at each member, which tests both predicted transitions and assumption (M) in one beam time.

References

1. K. O. Ho, C. Dailledouze, V. Žalandauskas, *et al.*, "Optical Stability and Photophysics of NV Centers in Diamond up to 120 GPa," arXiv:2606.02399 (2026).
2. A. Dréau *et al.*, "Avoiding power broadening in optically detected magnetic resonance of single NV defects," Phys. Rev. B **84**, 195204 (2011).
3. P. Bhattacharyya, *Principle and Applications of Quantum Metrology in Diamond Anvil Cells using the Nitrogen Vacancy Color Center*, Ph.D. thesis, UC Berkeley (Fall 2022), Ch. 6 and p. 65.
4. Repository: `code/ladder_detection.py`, `code/theory_a2_multiplicity.py`, `code/fig10_ladder_detection.py` (this analysis); `theory_freeze_v3_ho_integrated.md` (A2, N1–N4), `STATUS.md` §6.